

Dynamic Mode Decomposition

Aim: Learn linear models from snapshot data.

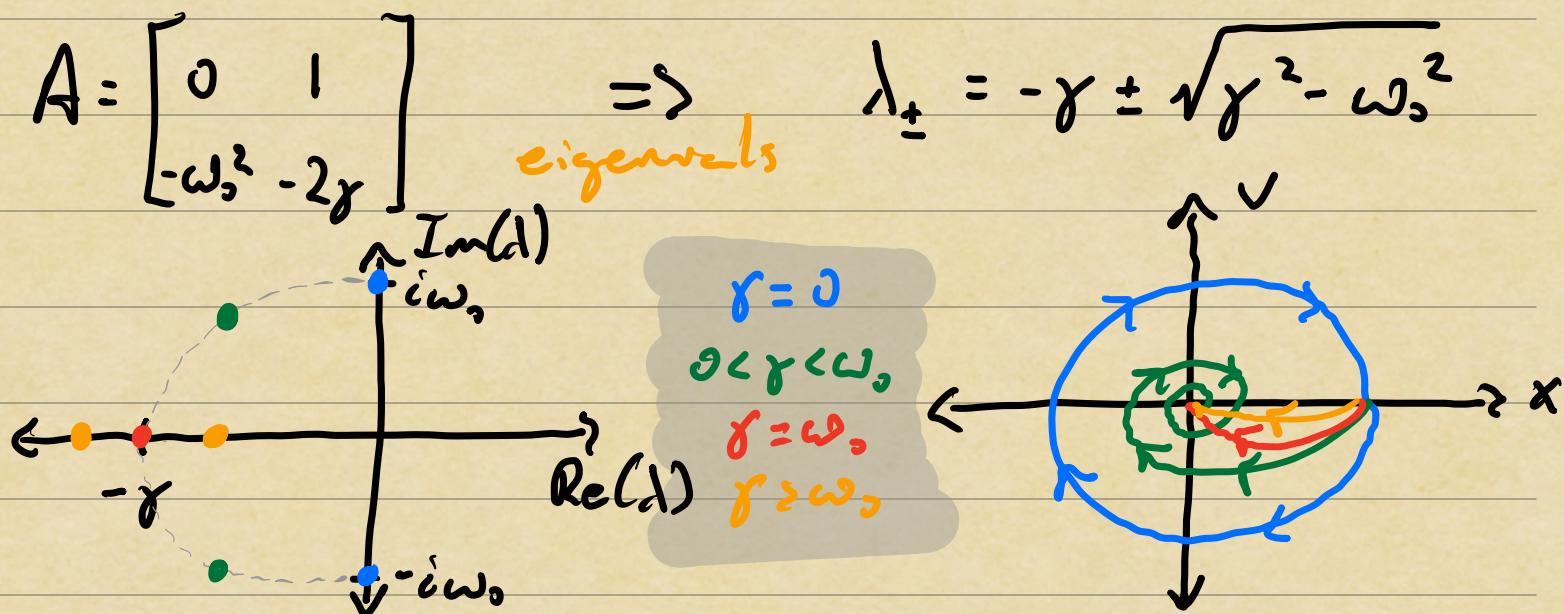
~~Recall~~ A linear dynamized system has form

$$(*) \quad \dot{x} = Ax, \quad x(0) = x_0.$$

The solution is the matrix exponential

$$x(t) = \exp(At)x_0, \quad t \geq 0.$$

Example. A simple oscillator.



Discrete-Time Linear Systems ("Snapshot" Dynamics)

Suppose that we observe trajectories of the dynamical system (*) sampled at regular intervals $t_0 = 0, t_1 = \Delta t, \dots, t_m = m \Delta t$.

$$X = \begin{bmatrix} 1 & & 1 \\ x_0 & \dots & x_{m-1} \\ 1 & & 1 \end{bmatrix}, \quad Y = \begin{bmatrix} 1 & & 1 \\ x_1 & \dots & x_m \\ 1 & & 1 \end{bmatrix}$$

where $x_{j+1} = \exp(A \Delta t) x_j$, $j = 0, \dots, m-1$.

In its simplest form, the Dynamic Mode Decomposition aims to recover the matrix

$$T_{\Delta t} = \exp(A \Delta t)$$

from the snapshot pairs in X and Y .

Q: How can we learn a matrix from "snapshots," e.g., input-output pairs?

The simplest idea is to "fit" the snapshots:

$$(AX) \quad \tilde{T}_{\Delta t} = \underset{T \in \mathbb{R}^{d \times d}}{\operatorname{argmin}} \overset{d \times m \quad d \times d \quad d \times m}{\|Y - TX\|}.$$

$\tilde{T}_{\Delta t}$ can be used to make predictions, and we can look at the eigenvalues and eigenvectors to analyze the dynamics.

Q: What are the eigenvectors and eigenvalues of the matrix exponential?

$$\exp(A\Delta t) = \begin{bmatrix} 1 & & \\ & \ddots & \\ & & 1 \end{bmatrix} \begin{bmatrix} e^{\lambda_1 \Delta t} & & \\ & \ddots & \\ & & e^{\lambda_d \Delta t} \end{bmatrix} \begin{bmatrix} 1 & & \\ & \ddots & \\ & & 1 \end{bmatrix}^{-1}$$

$$\Rightarrow \exp(A\Delta t) u_k = \overset{\uparrow}{e^{\lambda_k \Delta t}} \overset{\uparrow}{u_k}$$

$\left[\begin{array}{c} \text{eigenvalue} \\ \text{eigenvector} \end{array} \right]$

The eigenvalues of the sampled dynamical system depend on the sampling rate Δt !

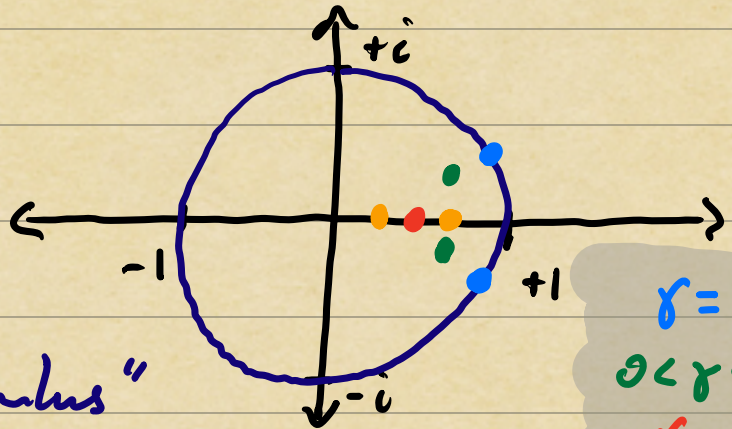
Example. Discrete-time oscillator.

$$x_j = \exp(A \Delta t) x_{j-1}, \quad j = 0, 1, 2, \dots$$

$$u = e^{\lambda \Delta t} = \rho e^{i\varphi \Delta t} \\ = \rho (\cos \varphi \Delta t + i \sin \varphi \Delta t)$$

$$\rho = e^{\Delta t \operatorname{Re}(\lambda)} = \text{"modulus"}$$

$$\varphi = \Delta t \operatorname{Im}(\lambda) = \text{"argument"}$$



$\gamma = 0$
 $0 < \gamma < \omega_0$
 $\gamma = \omega_0$
 $\gamma > \omega_0$

Q: How well does DM1D recover the exponential and its eigenvalues from a sampled trajectory of the oscillator?

$\Rightarrow$ See demo 03.m on Github.

Q: Is it possible to recover $T_{\Delta t}$ exactly? How many snapshots are needed? What if the data is noisy?

DMD and Least-Squares

For low-dimensional linear dynamics like the oscillator, we often have $d \ll m$.

If the 2-norm is used in (**), then $\tilde{\Gamma}_{\Delta t}^T$ solves the overdetermined LS-problem:

$$\tilde{\Gamma}_{\Delta t}^T = \underset{S \in \mathbb{R}^{d \times d}}{\operatorname{argmin}} \left\| \begin{bmatrix} -x_0 - \\ \vdots \\ -x_{m-1} - \end{bmatrix} [S] - \begin{bmatrix} -x_1 - \\ \vdots \\ -x_m - \end{bmatrix} \right\|_2$$

To analyze the accuracy & sensitivity of our "learned" models, we can apply the same tools we used for model regression earlier (here, the model is linear but multivariate).

$\Rightarrow$ How well is the ground truth approximated by a linear map?

$\Rightarrow$ How sensitive is the linear fit to data?

The
2-key
questions!

DMD is often applied to dynamical systems with high-dimensional state-space and relatively small number of snapshots. Here, $m \ll d$ and the LS-problem is underdetermined. There is no unique soln!

$$\underset{s \in \mathbb{R}^d}{\operatorname{argmin}} \left\| \begin{bmatrix} \text{---} x_0 \text{---} \\ \vdots \\ \text{---} x_{m-1} \text{---} \end{bmatrix} \begin{bmatrix} s \end{bmatrix} - \begin{bmatrix} \text{---} x_1 \text{---} \\ \vdots \\ \text{---} x_m \text{---} \end{bmatrix} \right\|$$

If $\operatorname{rank}(X) = m$, ∞ -ly many solutions.

Q: Which soln should we pick?